Higher-Order Asymptotics of Regression-Adjusted Estimators in Randomized Trials*

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Abstract

Regression adjustment is widely used to improve the precision of average treatment effect estimates from randomized controlled trials. In the important case of equally balanced trials, simple regression and interacted regression have the same first-order variance, meaning the optimal estimator is determined by higher-order terms. We derive a closed-form expression for the second order difference in their mean squared error under complete randomization and superpopulation sampling, and show that this difference, divided by the common first-order variance, is the asymptotic effective sample size loss from using the less efficient estimator. We decompose the expression into interpretable effects from covariate shape, treatment-effect heterogeneity, heteroskedasticity, and nonlinear misspecification, and show that it can favor either estimator. However, we demonstrate that in settings typical of industry data—in particular, heavy-tailed and skewed covariates with heteroskedastic deviations from the linear model—interacted regression is actually less precise, and we quantify this gap over a range of data generating processes.

1 Introduction

Randomized controlled trials (RCTs) are fundamental to how causal effects are ascertained across a number of fields. One valuable attribute of RCTs is that they enable unbiased estimation of average treatment effects using simple methods such as the difference in means, without reliance on parametric assumptions [Imbens and Rubin, 2015]. However, while the difference-in-means estimator is unbiased, this does not automatically make it the preferred approach; and in particular regression adjustments are often used to improve power. For example, the Food and Drug Administration issued recent guidance encouraging analysts to "adjust for baseline covariates that are strongly associated with an outcome in a clinical trial population" [U.S. Food and Drug Administration, 2023, accompanying guidance snapshot].

The study of regression adjustments in RCTs has a long tradition in statistics. Fisher [1935] used arguments rooted in parametric linear modeling to advocate for such regression adjustments. Fisher's analysis was later critiqued by Freedman [2008], who argued that reliance on linear models was misaligned with model-free guarantees sought in RCT inference. Lin [2013], however, responded to Freedman's critique by showing that regression adjustments do not require parametric assumptions to be near-unbiased for the average treatment effect.

*Code, simulation scripts and the computational certificates used in the proofs: https://github.com/kelleyan-pixel/higher-order-regression-adjustment-for-rct.

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Furthermore, Lin [2013] showed that regression adjustments with full treatment-control interaction weakly dominate both the difference in means estimator and Fisher's simple regression adjustment in terms of their leading-order (root-n scale) errors. Lin's weak dominance result has led such treatment-interacted regressions to be often presented as a safe starting point for use of regression adjustments in RCTs [Li and Ding, 2017, Negi and Wooldridge, 2021, Wager, 2025].

A gap in this narrative, however, is that treatment-interacted regressions only strictly dominate simple regression in designs with randomization probability $\pi \neq 0.5$ (and treatment effects that vary with the covariates); the dominance is weak at every π. In evenly randomized trials with $\pi = 0.5$, the first-order accuracy of both methods is equal for any data-generating distribution, i.e., first-order theory offers no guidance on which method to prefer. And, while $\pi = 0.5$ is of course a special case, this is an important special case in applications, and it is not a priori clear we should prefer interacted regression to simple regression here. For example, Zhao and Ding [2023] argue based on simulation evidence and heuristics on the number of free parameters that simple regression may in practice be more stable than interacted regression with limited sample size—especially when there are multiple treatment levels.

Indeed, first-order equivalence can easily break down in finite sample sizes, as in the following illustrative example. Suppose a tech company runs an A/B test on a sample of its user base and adjusts for X_i, each user's spend in the prior time period. X is strongly right-skewed because it is bounded below and a small number of heavy users account for the majority of the total. Suppose also that these superusers have more volatile outcomes, so that outcome noise increases with X_i, and that the treatment shifts every user's outcome by the same amount. Specifically, let the raw X_i be marginally LogNormal$(0, 1.1^2)$ and let

$$\sigma(X_i) = 1 + X_i, \qquad Y_i(0) = 0.9\, X_i + \sigma(X_i) u_i, \qquad Y_i(1) = Y_i(0) + 1,$$

with $u_i \overset{\text{iid}}{\sim} N(0,1)$. In simulation, interacted regression's MSE exceeds that of simple regression by 6.7% at $n = 500$, and asymptotically, the former estimator requires an additional sample size of 83 to catch up to the latter in precision. The asymptotic figure should be read as a guide to sign and rough magnitude rather than an exact prediction at moderate n: here the second-order approximation implies an excess MSE of 16.7% at $n = 500$, against the simulated 6.7% (Appendix C).

Motivated by this gap, we here conduct a higher-order asymptotic analysis of both simple and interacted regression adjustments in RCTs to elucidate relative merits of both approaches in the case where $\pi = 0.5$. We frame our findings in terms of asymptotic deficiency [Hodges and Lehmann, 1970]. When two estimators have matching first-order behavior but one dominates the other on the second-order scale then, asymptotically, switching to the better one gives an accuracy boost that is equivalent to a fixed boost in sample size, which is referred to as the deficiency. Our main finding is that, while either the simple or interacted regression adjustment may have better second-order performance, this trade-off appears to favor simple regression in realistic settings.

As a practical extension of this theory, we study feasible inference choices for both estimators with heteroskedasticity-robust standard error estimates. In particular, we consider both interacted and simple regression point estimates in tandem with HC1 and jackknife standard error estimates. For both point estimates, HC1 intervals are miscalibrated under skewed, tail-heteroskedastic covariates: in our low-heterogeneity designs they are close to nominal at the 95% level but undercover at lower levels, down to 75.1% at nominal 80%. The jackknife is at or above nominal coverage, up to Monte Carlo error, in all of our low-heterogeneity designs, conservatively so under the heaviest tails, but it still undercovers when large treatment-effect

heterogeneity meets skewed covariates. In a typical low-HTE regime, simple regression combined with the jackknife achieves nominal coverage, up to Monte Carlo error, at every level we examine, with intervals at most 0.5% wider, and up to 14% narrower, than those of interacted regression with either standard error.

1.1 Related Work

Regression adjustment in randomized experiments. The literature on regression adjustment in randomized experiments sits in two camps. The design-based literature treats the potential outcomes and covariates as fixed in a finite population, with randomness coming solely from treatment assignment. In this setting, Lin [2013] shows that interacted regression weakly dominates both simple regression and difference-in-means in asymptotic efficiency. Meanwhile, Zhao and Ding [2023] compare their finite sample behavior and allow for multi-armed and factorial experiments, postulating that interacted regression exhibits poor finite sample performance due to its additional model complexity. Another interesting but more distant line of work, including Lei and Ding [2021] and Chiang et al. [2025], addresses bias and inference of the interacted estimator as p grows with n.

By contrast, the superpopulation framework treats the experimental units as random draws from a larger population and studies repeated sampling over both the units and treatment assignment. Most past work here has focused on first-order comparisons. For instance, Negi and Wooldridge [2021] show that the interacted regression estimator weakly dominates both difference in means and simple regression in this framework as well. We continue this line of work to formally investigate higher-order effects of both estimators, in particular to resolve the ambiguous optimality at $\pi = 0.5$. Importantly, our goal is to characterize the higher-order behavior of the two estimators under IID sampling specifically, and it is possible that a design-based analysis yields different second-order results.

Higher-order asymptotic theory. First-order theory inherently cannot compare estimators with the same asymptotic variance, even though their finite-sample behavior may differ meaningfully. To address this, Nagar [1959] introduced expansions of the higher-order moments of estimators, and Sargan [1974] gave conditions for their validity; meanwhile Edgeworth expansions approximate their underlying distribution [Sargan, 1976, Bhattacharya and Rao, 1976, Hall, 1992, Ghosh, 1994]. As we focus on the mean-squared error, we primarily build on the former tools, although the latter are useful for rigorous results on tail behavior and inference. Hodges and Lehmann [1970] introduced asymptotic deficiency as a measure of the additional sample size required by the less accurate of two first-order equivalent procedures, and Rothenberg [1984] emphasized that higher-order approximations should be consulted when competing estimators have the same limiting distribution. Parallel to our contribution, Newey and Smith [2004] present a higher-order comparison of the first-order equivalent general method of moments and generalized empirical likelihood estimators. Tan [2014] studies the second-order properties of regression and likelihood calibration estimators in survey sampling and missing-data problems, finding that neither consistently dominates. Chiang et al. [2025] derive second-order stochastic expansions of regression-adjusted estimators to compare inference in a design-based, growing p regime. However, to our knowledge we are the first to compare interacted and simple regression in a superpopulation setting using higher-order expansions.

Variance estimation. Under superpopulation sampling, the fixed-design based conventional sandwich variance for the interacted estimator ignores the variance of the sample covariate mean used for centering [Imbens and Wooldridge, 2009, Section 5.3]; see also Lin [2013, Section 5].

Imbens and Wooldridge [2009] add an estimate of the omitted term, and Negi and Wooldridge [2021] derive the asymptotic variance that accounts for it; we take this result for granted. General finite-sample behavior of heteroskedasticity-robust variance estimators is studied by MacKinnon and White [1985], Chesher and Jewitt [1987], Long and Ervin [2000] and Imbens and Kolesár [2016]. It is well known that despite their popularity in applied research, HC0 and HC1 are prone to undercovering in finite samples, and the statistics literature typically recommends switching to HC3. Lei and Ding [2021] and Chiang et al. [2025] demonstrate this point specifically in the case of regression adjustment inference. Following this work, we present an empirical study of feasible inference using regression adjustment combined with robust standard errors, but we do not claim to contribute to this literature from a theoretical standpoint.

2 Second-Order Analysis

We focus on completely randomized trials (CRTs) with causal effects formalized using the potential outcomes model [Imbens and Rubin, 2015]. We assume the covariates $X_i \in \mathbb{R}^p$ and potential outcomes $Y_i(0), Y_i(1) \in \mathbb{R}$ are drawn IID from a distribution P, with average treatment effect $\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$. Both regression estimators below include an intercept, and interacted regression is evaluated at the sample mean of the covariates, so translating the covariates leaves both estimators unchanged; we therefore assume without loss of generality that $\mathbb{E}[X] = 0$. Treatment is assigned randomly conditional on the fixed treated group size $n_1 = \sum_{i=1}^n W_i = \lfloor \pi n \rfloor$; treatment assignments are exchangeable but not independent. The analyst observes (X_i, Y_i, W_i) for $i = 1, \ldots, n$, where $Y_i = Y_i(W_i)$. We consider the following three estimators of the ATE:

$$\begin{aligned}
\hat{\tau}_{DM} &= \frac{1}{|\{i : W_i = 1\}|} \sum_{\{i:W_i=1\}} Y_i - \frac{1}{|\{i : W_i = 0\}|} \sum_{\{i:W_i=0\}} Y_i, \\
\hat{\tau}_{REG} &= \hat{\tau}, \text{ with } Y_i \sim \alpha + X_i\beta + W_i\tau, \\
\hat{\tau}_{IREG} &= \hat{\tau} + \frac{1}{n}\sum_{i=1}^n X_i\hat{\delta}, \text{ with } Y_i \sim \alpha + X_i\beta + W_i\tau + W_iX_i\delta,
\end{aligned} \tag{1}$$

where, e.g., $Y_i \sim \alpha + X_i\beta + W_i\tau$ means we algorithmically run a regression of Y_i against X_i and W_i with an intercept (without making any claims on well-specification of such a regression). To describe their first-order behavior it is helpful to define the best linear approximation to the conditional response function of the potential outcomes,

$$\left\{\alpha^*_{(w)},\ \beta^*_{(w)}\right\} = \underset{\alpha_{(w)},\,\beta_{(w)}}{\operatorname{argmin}} \left\{\mathbb{E}\left[\left(Y_i(w) - X_i\beta_{(w)} - \alpha_{(w)}\right)^2\right]\right\}, \tag{2}$$

together with the oracle residuals and the scaled heterogeneity direction

$$\begin{aligned}
\varepsilon_i(w) &= Y_i(w) - \alpha^*_{(w)} - X_i^\top\beta^*_{(w)}, \\
Z_i = \Sigma^{-1/2}X_i, \quad \Sigma = \operatorname{Var}[X_i], &\qquad \gamma = \Sigma^{1/2}\left(\beta^*_{(1)} - \beta^*_{(0)}\right),
\end{aligned} \tag{3}$$

so that $\mathbb{E}[Z_i] = 0$, $\mathbb{E}[Z_iZ_i^\top] = I_p$, $\mathbb{E}[Z_i\varepsilon_i(w)] = 0$ and $\varepsilon_i = \varepsilon_i(W_i)$. All population quantities below are invariant to the choice of matrix square root: replacing $\Sigma^{1/2}$ by $\Sigma^{1/2}O$ for an orthogonal O maps $Z \mapsto O^\top Z$ and $\gamma \mapsto O^\top\gamma$, which leaves every scalar in Theorem 2 unchanged. The first-order behavior of the three estimators is well known [Negi and Wooldridge, 2021].

Proposition 1. *Suppose that X_i and $Y_i(w)$ have finite second moments and that* $\mathrm{Var}[X_i]$ *is invertible. Then all three estimators above are asymptotically normal,* $\sqrt{n}(\hat{\tau}_\cdot - \tau) \Rightarrow \mathcal{N}(0, V_\cdot)$*, with limiting variances*

$$
\begin{aligned}
V_{DM} &= V_{IREG} + \mathrm{Var}\left[X_i\left(\sqrt{\frac{\pi}{1-\pi}}\,\beta^*_{(0)} + \sqrt{\frac{1-\pi}{\pi}}\,\beta^*_{(1)}\right)\right], \\
V_{REG} &= V_{IREG} + \frac{(1-2\pi)^2}{\pi(1-\pi)}\,\mathrm{Var}\left[X_i\left(\beta^*_{(1)} - \beta^*_{(0)}\right)\right], \\
V_{IREG} &= \mathrm{Var}\left[X_i\left(\beta^*_{(1)} - \beta^*_{(0)}\right)\right] + \sum_{w=0,\,1} \frac{1}{\mathbb{P}\left[W_i = w\right]}\mathbb{E}\left[\left(Y_i(w) - X_i\beta^*_{(w)} - \alpha^*_{(w)}\right)^2\right].
\end{aligned}
\tag{4}
$$

Thus $\hat{\tau}_{IREG}$ weakly dominates both $\hat{\tau}_{DM}$ and $\hat{\tau}_{REG}$ at every π; its advantage over $\hat{\tau}_{REG}$ is strict for generic distributions P when $\pi \neq 0.5$, while at $\pi = 0.5$ the two regression estimators have the same asymptotic variance

$$
V^* = \|\gamma\|^2 + 4\mathbb{E}\left[\varepsilon_i^2\right] = \|\gamma\|^2 + 2\mathbb{E}\left[\varepsilon_i(1)^2\right] + 2\mathbb{E}\left[\varepsilon_i(0)^2\right]. \tag{5}
$$

This is the setting we study. Because first-order theory is indifferent between the two methods, we turn to second-order asymptotics.

Suppose we compare two estimators $\hat{\tau}_A$ and $\hat{\tau}_B$ of a shared quantity τ whose mean-squared errors agree on the (first-order) $1/n$ but not on the (second-order) $1/n^2$ scale,

$$
\mathrm{MSE}(\hat{\tau}_A) = \frac{V}{n} + \frac{c_A}{n^2} + o\left(\frac{1}{n^2}\right), \qquad \mathrm{MSE}(\hat{\tau}_B) = \frac{V}{n} + \frac{c_B}{n^2} + o\left(\frac{1}{n^2}\right). \tag{6}
$$

Hodges and Lehmann [1970] propose quantifying the difference in performance by the number of extra samples required to compensate for using the less accurate estimator: they solve for k_n such that $\hat{\tau}_B$ run on k_n samples matches the accuracy of $\hat{\tau}_A$ on n samples as closely as possible. When $\lim_{n\to\infty}(k_n - n) = \Delta \in \mathbb{R}$ exists, Δ is the asymptotic deficiency of $\hat{\tau}_B$ relative to $\hat{\tau}_A$. Under (6),

$$
\Delta = (c_B - c_A)\,/\,V. \tag{7}
$$

Our work derives the second order stochastic expansion of the MSE. Importantly, the true MSE may not converge as $n \to \infty$ if the estimators have divergent higher moments. Under the additional regularity conditions that the estimators are well defined and that the Taylor remainder is negligible in mean square at the n^{-2} scale, the "expansion MSE" that we derive agrees with the usual finite-sample MSE up to $o(n^{-2})$.

Definition 1 (Expansion MSE)**.** Let $\mathcal{V}_i$ collect the per-unit moment coordinates entering the coefficient function of an estimator, let $\mu = \mathbb{E}[\mathcal{V}_i]$, $\bar{\mathcal{V}} = n^{-1}\sum_i \mathcal{V}_i$, and suppose $\hat{\tau} = g(\bar{\mathcal{V}})$ for a function g that is three times continuously differentiable in a neighborhood of μ with $g(\mu) = \tau$. Write $\zeta^{(n)} = \sqrt{n}(\bar{\mathcal{V}} - \mu)$ and let

$$
\tilde{\tau}_n = \tau + \frac{1}{\sqrt{n}} a_i \zeta_i^{(n)} + \frac{1}{2n} H_{ij}\zeta_i^{(n)}\zeta_j^{(n)} + \frac{1}{6n^{3/2}} A_{3,ijk}\zeta_i^{(n)}\zeta_j^{(n)}\zeta_k^{(n)} \tag{8}
$$

be the third-order Taylor polynomial of g at μ, with a, H and A_3 the first three derivative tensors of g at μ. Our object of interest is $\mathrm{MSE}^{\mathrm{exp}}(\hat{\tau}) = \mathbb{E}[(\tilde{\tau}_n - \tau)^2]$.

Furthermore, we make the following assumption so that $\mathrm{MSE}^{\mathrm{exp}}$ is well-defined.

Assumption 1 (Sampling and moments)**.** $(X_i, Y_i(0), Y_i(1))$ are IID from P, $\Sigma = \mathrm{Var}[X_i] \succ 0$, p is fixed, $\pi = 1/2$ with $n_1 = n/2$ for even n (and $n_1 = \lfloor n/2 \rfloor$ otherwise), and

$$\mathbb{E}\big[\|X_i\|^{12}\big] + \sum_{w=0}^{1} \mathbb{E}\big[|Y_i(w)|^{12}\big] < \infty.$$

Note that this implies $\mathbb{E}\|\mathcal{V}_i\|^6 < \infty$ by the Cauchy–Schwarz inequality.

2.1 The second-order difference

Theorem 2. *Let Assumption 1 hold, and define*

$$K = \mathbb{E}[ZZ^\top \|Z\|^2], \qquad S = \mathbb{E}[Z\|Z\|^2], \qquad S_\gamma = \mathbb{E}[ZZ^\top (Z^\top \gamma)],$$

$$\mathcal{E} = \mathbb{E}[ZZ^\top \varepsilon], \qquad \mathcal{E}_W = \mathbb{E}[ZZ^\top (W - \tfrac{1}{2})\varepsilon] = \tfrac{1}{4}\big(\mathbb{E}[ZZ^\top \varepsilon(1)] - \mathbb{E}[ZZ^\top \varepsilon(0)]\big).$$

With $\mathrm{MSE}^{\mathrm{exp}}$ *as in Definition 1 and* W *assigned via balanced CRT,*

$$\mathrm{MSE}^{\mathrm{exp}}(\hat{\tau}_{\mathrm{IREG}}) - \mathrm{MSE}^{\mathrm{exp}}(\hat{\tau}_{\mathrm{REG}}) = \frac{D_p}{n^2} + O(n^{-3}),$$

where

$$\begin{aligned} D_p = & -(2p+3)\|\gamma\|^2 + \gamma^\top K \gamma - (\gamma^\top S)^2 - 3\|S_\gamma\|_F^2 \\ & - 16\,\mathrm{tr}(S_\gamma \mathcal{E}_W) - 8(\gamma^\top S)\,\mathrm{tr}(\mathcal{E}_W) + 8\gamma^\top \mathcal{E}_W S + 8\,\mathrm{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2]. \end{aligned}$$

Consequently the asymptotic deficiency of IREG relative to REG, computed from the expansion MSE, is $\Delta = D_p / V^*$ *with* V^* *as in* (5).

The proof is in Appendix A with numerical verification and demonstration in Appendix B.

The expression for D_p combines four key blocks: a covariate shape block scaled by the linear HTE γ; terms crossing γ with residual asymmetry; residual heteroskedasticity; and nonlinear misspecification. No single term determines the sign; the terms matter through their signs and magnitudes. For instance, when $\gamma = 0$ all terms involving γ vanish and

$$D_p = 8\,\mathrm{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2].$$

While $\mathrm{tr}(\mathcal{E}^2) = \|\mathcal{E}\|_F^2 \geq 0$, the term $S^\top \mathbb{E}[Z\varepsilon^2]$ can have either sign, so Δ can be positive or negative in this case.

Remark 3 (Bias and variance decomposition). One may naturally be interested in how the overall MSE difference decomposes into bias and variance. The bias for estimator $\tilde{\tau}_n$ is $\mathbb{E}[\tilde{\tau}_n - \tau] = \frac{1}{2}\,\mathrm{tr}(H\Sigma_n) + O(n^{-2})$, where H is the Hessian of the estimator's coefficient function and Σ_n is the covariance of the per-arm means. Evaluating this leading term for the two estimators (Lemma 13 in Appendix A) gives

$$\mathbb{E}[\tilde{\tau}_{n,\mathrm{REG}} - \tau] = -\frac{4\,\mathrm{tr}(\mathcal{E}_W) + \gamma^\top S}{n} + O(n^{-2}), \qquad \mathbb{E}[\tilde{\tau}_{n,\mathrm{IREG}} - \tau] = -\frac{4\,\mathrm{tr}(\mathcal{E}_W)}{n} + O(n^{-2}),$$

so that $n^2\{\mathrm{Bias}^2_{\mathrm{IREG}} - \mathrm{Bias}^2_{\mathrm{REG}}\} = -(\gamma^\top S)^2 - 8(\gamma^\top S)\,\mathrm{tr}(\mathcal{E}_W)$ to leading order, indeed directly reproducing two terms; the remaining terms are due to variance.

To interpret the scale of Δ we define the variance share due to HTE as

$$R_\tau^2 = \frac{\|\gamma\|^2}{\|\gamma\|^2 + 4\mathbb{E}[\varepsilon_i^2]} = \frac{\|\gamma\|^2}{V^*}.$$

Consider the balanced doubly robust pseudo-outcome built from interacted linear model $\mu_w(x) = \alpha^*_{(w)} + x^\top \beta^*_{(w)}$,

$$\phi_i = 2W_i\{Y_i - \mu_1(X_i)\} - 2(1 - W_i)\{Y_i - \mu_0(X_i)\} + \mu_1(X_i) - \mu_0(X_i).$$

Then $\phi_i - \tau = \gamma^\top Z_i + \xi_i$ with $\xi_i = 2W_i\varepsilon_i(1) - 2(1 - W_i)\varepsilon_i(0)$; the normal equations give $\mathbb{E}[Z_i\xi_i] = 0$, $\mathrm{Var}[\xi_i] = 4\mathbb{E}[\varepsilon_i^2]$ and $\mathrm{Var}[\gamma^\top Z_i] = \|\gamma\|^2$, so R_τ^2 is the population R^2 from linearly predicting the treatment-effect pseudo-outcome from Z_i.

Corollary 4. *In addition to Assumption 1, suppose the potential outcomes satisfy a well-specified linear model with homoskedastic errors, $\mathbb{E}[\varepsilon_i(w) \mid X_i] = 0$ and $\mathbb{E}[\varepsilon_i^2(w) \mid X_i] = \sigma_w^2$ for $w \in \{0, 1\}$. Then $\mathcal{E} = \mathcal{E}_W = 0$ and $\mathbb{E}[Z\varepsilon^2] = 0$. For $\gamma \neq 0$,*

$$\Delta = R_\tau^2 \left[-(2p + 3) + \frac{\gamma^\top K\gamma - (\gamma^\top S)^2 - 3\|S_\gamma\|_F^2}{\|\gamma\|^2} \right]. \tag{9}$$

In the Gaussian sub-case $K = (p + 2)I_p$, $S = 0$ and $S_\gamma = 0$, so that $\Delta = -(p + 1)R_\tau^2$. For $\gamma = 0$, the corresponding expression follows directly from Theorem 2.

In the Gaussian homoskedastic case, the second-order coefficient favors IREG and grows linearly in the number of covariates, with p fixed as $n \to \infty$. This complicates the parameter-count heuristic of Zhao and Ding [2023], showing that the finite sample advantage of either estimator cannot be reduced to any simple feature of the DGP. However, the following corollary explains why we may expect IREG to perform worse in many real-life datasets.

Corollary 5. *In addition to Assumption 1, suppose the potential outcomes follow*

$$Y_i(w) = \alpha^*_{(w)} + X_i^\top \beta^*_{(w)} + \eta_i\|Z_i\|,$$

where $\mathbb{E}[\eta_i] = 0$, $\mathbb{E}[\eta_i^2] = \nu^2$ and η_i is independent of (Z_i, W_i). Then $\mathbb{E}[\varepsilon_i(w) \mid Z_i] = 0$ and $\mathbb{E}[\varepsilon_i^2(w) \mid Z_i] = \nu^2\|Z_i\|^2$, where $\|\cdot\|$ is the Euclidean norm, so that $\mathbb{E}[Z_i\varepsilon_i^2(w)] = \nu^2 S$ and $\mathbb{E}[\varepsilon_i^2] = \nu^2 p$. For $\gamma \neq 0$,

$$\Delta = R_\tau^2 \left[-(2p + 3) + \frac{\gamma^\top K\gamma - (\gamma^\top S)^2 - 3\|S_\gamma\|_F^2}{\|\gamma\|^2} \right] + \frac{8\nu^2 S^\top S}{\|\gamma\|^2 + 4\nu^2 p}. \tag{10}$$

For $\gamma = 0$, the corresponding expression again follows directly from Theorem 2.

Interpretation and scaling. It is useful to consider how the terms in (10) compare in practice. In many applications the linear HTE signal is small relative to the residual departure from the regression model, so R_τ^2 is small and the first summand rarely dominates unless its bracket is large. The bracketed component is of order p in the leading cases—it equals $-(p+1)$ under Gaussian covariates—and its moment terms grow with the kurtosis and skewness of the underlying covariates, so it is sensitive to departures from Gaussianity in either direction.

The remaining term is nonnegative, since $8\nu^2\|S\|^2/(\|\gamma\|^2 + 4\nu^2 p) \geq 0$, and for $\nu > 0$ it vanishes if and only if $S = \mathbb{E}[Z\|Z\|^2] = 0$. Right-skewed covariates with a long upper tail, such as measures of user activity, engagement or spending, can produce a large third-moment vector

S after centering and whitening, in which case this term can dominate the first summand. The scalar case makes this transparent: with $p = 1$, Corollary 5 gives

$$\Delta = R_\tau^2 \left\{ \mathbb{E}\left[Z_i^4\right] - 4\mathbb{E}\left[Z_i^3\right]^2 - 5 \right\} + \frac{8\nu^2 \mathbb{E}\left[Z_i^3\right]^2}{\gamma^2 + 4\nu^2}.$$

Using Pearson's inequality $\mathbb{E}\left[Z_i^4\right] \geq \mathbb{E}\left[Z_i^3\right]^2 + 1$ [see, e.g., Rohatgi and Székely, 1989] and $8\nu^2/(\gamma^2 + 4\nu^2) = 2(1 - R_\tau^2)$,

$$\Delta \geq -4R_\tau^2 + (2 - 5R_\tau^2)\mathbb{E}\left[Z_i^3\right]^2.$$

So when $R_\tau^2 < 2/5$ the deficiency is positive as soon as $\mathbb{E}\left[Z_i^3\right]^2 > 4R_\tau^2/(2 - 5R_\tau^2)$, and the lower bound grows linearly in $\mathbb{E}\left[Z_i^3\right]^2$.

3 Feasible Inference

While the above results concern the precision of the point estimates provided by REG and IREG, in practice we are also interested in the validity of the resulting confidence intervals—which now depends jointly on how we estimate the standard error. As before, we operate in an assumption-lean framework, only considering estimators that are robust to misspecification of the underlying linear model.

3.1 Implementation and variance estimators

We fit REG and IREG as in the R package `estimatr` [Blair et al., 2024]: REG is `lm_robust(y ~ w + x)`, and IREG is `lm_lin(y ~ w, covariates = ~ x)`, with `se_type = "HC1"` requested explicitly (the package default is HC2). For speed, the simulations below use a vectorized NumPy implementation of the same estimators, which reproduces `estimatr`'s estimates and standard errors; the replication repository documents the cross-check. Because we operate in a superpopulation framework while these implementations are for fixed design sampling, we add a plug-in estimate of the variance of the sample covariate mean used for centering (Imbens and Wooldridge, 2009, Section 5.3; Negi and Wooldridge, 2021):

$$\hat{G} = \hat{\delta}^\top \hat{\Sigma}_X \hat{\delta}, \qquad \widehat{\mathrm{Var}}_{\mathrm{SP}}(\hat{\tau}_{\mathrm{IREG}}) = \widehat{\mathrm{Var}}_{\mathrm{HC1}}(\hat{\tau}_{\mathrm{IREG}}) + \frac{\hat{G}}{n}, \tag{11}$$

where $\hat{\delta}$ is the vector of interaction coefficients and $\hat{\Sigma}_X$ the sample covariance of the covariates; $\hat{G}$ estimates $G = \|\gamma\|^2$. Because $\hat{\delta}$ is itself noisy, $\hat{G}$ is biased upward, by about $\mathrm{tr}\{\mathrm{Var}(\hat{\delta})\,\Sigma\}$.

We compare two choices of robust standard error estimates. HC1 is a popular benchmark: it is Stata's default and used in 98.1% of 40,571 robust Stata regressions in reproduction packages of leading economics journals [Kranz, 2024]. We compare it with a leverage-aware alternative, the delete-one jackknife (JK) applied to the final estimate [MacKinnon and White, 1985]. Let $\hat{\tau}_{(-i)}$ be the estimator recomputed on the sample with unit i removed—both arm-specific fits are refitted and, for IREG, the covariates are recentred at the leave-one-out mean $\bar{X}_{(-i)}$—the JK variance estimate is then given by

$$\hat{V}_{\mathrm{JK}} = \frac{n-1}{n} \sum_{i=1}^{n} \left(\hat{\tau}_{(-i)} - \bar{\hat{\tau}}\right)^2, \qquad \bar{\hat{\tau}} = \frac{1}{n} \sum_{i=1}^{n} \hat{\tau}_{(-i)}. \tag{12}$$

Because JK operates on the final treatment effect estimate, including the recentring at $\bar{X}$, it accounts for the sampling variation of the covariate mean and thus requires no superpopulation correction. Thus for IREG we report HC1 with the plug-in correction (11) alongside the jackknife; for REG, we directly report HC1 and JK. For least squares, JK is closely approximated by HC3 [MacKinnon and White, 1985]. We compute $\hat{\tau}_{(-i)}$ with the exact deletion formula for least squares, which agrees with refitting to machine precision. When removing unit i leaves a rank-deficient design (unit i has leverage one in its fit), $\hat{\tau}_{(-i)}$ is undefined; we then omit it and apply (12) to the remaining n' leave-one-out estimates, with n' in place of n. All intervals use t critical values with $n-k$ degrees of freedom, where k is the number of regression coefficients, and a replication without a finite interval counts as a non-cover.

3.2 Numerical evidence

Table 1 compares all four joint procedure possibilities across 14 DGPs. The selection of DGPs we analyze is deliberately diverse, with a range of R^2_τ and covariates—Gaussian, skewed, heavy-tailed, and rare-value covariates; Appendix D defines each design. All have $n = 500$ and 20,000 replications, and every procedure is applied to the same simulated datasets.

Table 1: Root-mean-squared error of the point estimators, and median width and coverage (%) of nominal 95% intervals for the population ATE: IREG with HC1 plus the plug-in correction and with the direct jackknife (12), and REG with HC1 and with the direct jackknife ($n = 500$, 20,000 replications; "het." denotes heteroskedastic noise). Designs are defined in Appendix D; g is the treatment-effect slope of the scalar lognormal designs. In the rare-value design, 6.3% of replications have a leverage-one unit in an IREG arm, whose leave-one-out estimate is omitted from the jackknife (Section 3.1).

		IREG					**REG**				
			HC1 + plug-in		jackknife			HC1		jackknife	
Design	R^2_τ	RMSE	width	cov.	width	cov.	RMSE	width	cov.	width	cov.
Small or no heterogeneity											
Gaussian	0.022	0.091	0.36	95.0	0.36	95.0	0.091	0.36	94.8	0.36	95.0
Misspecified, $G = 0$	0	0.098	0.38	94.9	0.38	95.0	0.098	0.38	94.8	0.38	95.0
Skewed, het., $G = 0$	0	0.167	0.65	95.0	0.66	95.4	0.165	0.64	95.0	0.65	95.5
Heavy-tailed, het.	0.016	0.175	0.64	94.7	0.66	95.5	0.168	0.63	94.7	0.64	95.5
Lognormal $s = 1.2$, $g = 0.3$	0.013	0.425	1.52	94.6	1.57	95.6	0.408	1.47	94.6	1.52	95.7
Lognormal $s = 1.5$, $g = 0.1$	0.002	0.798	2.55	94.7	2.73	96.2	0.715	2.40	94.9	2.55	96.7
Lognormal $s = 1.5$, $g = 0.5$	0.049	0.839	2.62	93.7	2.79	95.3	0.753	2.46	93.2	2.61	95.1
Lognormal $s = 1.8$, $g = 0.3$, $\sigma = 1 + 1.5X$	0.009	2.826	7.26	94.3	8.04	96.4	2.196	6.57	94.3	7.18	96.6
Lognormal $s = 2$, $g = 0.1$	0.002	3.433	7.76	94.8	8.78	97.0	2.458	6.80	95.0	7.55	97.6
Moderate heterogeneity											
Lognormal $s = 1.5$, $g = 1$	0.172	0.901	2.81	92.0	2.97	93.7	0.826	2.64	90.2	2.81	91.9
Rare extreme values	0.2	0.100	0.40	94.0	0.40	94.0	0.102	0.40	94.7	0.41	95.5
Large heterogeneity											
Heavy-tailed, het.	0.365	0.220	0.75	94.2	0.77	94.6	0.212	0.73	91.9	0.76	93.0
Lognormal $s = 1.5$, $g = 3$	0.651	1.451	4.46	89.8	4.57	90.4	1.380	4.17	84.1	4.51	86.0
Gaussian	0.692	0.161	0.64	95.1	0.64	95.1	0.162	0.64	95.0	0.64	95.2

When R^2_τ is small (< 0.05), the combination of REG and JK is the most attractive. Its 95% coverage ranges from 95.0% to 97.6%, its median interval width is between 0.86 and 1.00 times that of IREG with the jackknife and between 0.97 and 1.00 times that of IREG with HC1 plus the plug-in, which covers 93.7–95.0%. REG is also at least as precise a point estimator in these designs, with MSE ratios IREG/REG between 1.00 and 1.95. At the 95% level both estimators are close to nominal when paired with HC1 (REG+HC1 covers 93.2–95.0%); for

IREG this partly reflects the plug-in, whose upward bias offsets the understatement of HC1 at high-leverage points, and the agreement does not extend to other nominal levels (Table 2). The price of the jackknife is width: under the heaviest tails it is conservative, and REG+JK intervals are up to 1.11 times as wide as REG+HC1's.

This pattern does not hold across all DGPs. As R^2_τ grows with skewed covariates, REG undercovers with either variance estimator, down to 86.0% with JK at $R^2_\tau = 0.651$, while IREG with JK is the most reliable procedure, although it too reaches only 90.4% there. In the rare-value design an IREG arm contains a single rare unit in 6.3% of replications; removing that unit leaves a rank-deficient fit, so its leave-one-out estimate is omitted, and IREG+JK covers 89.9% of those replications against 94.3% of the others. Counting those replications as failures instead would lower IREG+JK coverage to 88.4%. REG is essentially unaffected: removing a unit leaves its pooled fit rank deficient in only 0.03% of replications.

Table 2: Coverage (%) at nominal 80%, 95% and 99% in the designs of Table 1 with small or no heterogeneity, for IREG with HC1 plus the plug-in and with the jackknife, and for REG with HC1 and with the jackknife.

		IREG, HC1 + plug-in			IREG, jackknife			REG, HC1			REG, jackknife		
Design	R^2_τ	80	95	99	80	95	99	80	95	99	80	95	99
Gaussian	0.022	80.1	95.0	99.1	80.2	95.0	99.1	79.9	94.8	99.0	80.2	95.0	99.0
Misspecified, $G = 0$	0	79.8	94.9	99.0	79.8	95.0	99.0	79.6	94.8	98.9	79.9	95.0	99.0
Skewed, het., $G = 0$	0	80.1	95.0	99.0	80.8	95.4	99.1	80.0	95.0	98.9	80.8	95.5	99.1
Heavy-tailed, het.	0.016	79.1	94.7	98.9	80.8	95.5	99.1	79.0	94.7	98.9	80.9	95.5	99.1
Lognormal $s = 1.2$, $g = 0.3$	0.013	78.4	94.6	98.9	80.6	95.6	99.2	78.6	94.6	98.9	80.8	95.7	99.2
Lognormal $s = 1.5$, $g = 0.1$	0.002	77.0	94.7	99.3	81.2	96.2	99.6	77.8	94.9	99.3	82.5	96.7	99.6
Lognormal $s = 1.5$, $g = 0.5$	0.049	76.6	93.7	98.5	80.4	95.3	98.9	76.4	93.2	98.3	80.6	95.1	98.8
Lognormal $s = 1.8$, $g = 0.3$, $\sigma = 1 + 1.5X$	0.009	75.5	94.3	99.1	81.5	96.4	99.5	76.4	94.3	99.0	83.1	96.6	99.5
Lognormal $s = 2$, $g = 0.1$	0.002	75.1	94.8	99.3	82.3	97.0	99.6	76.1	95.0	99.3	84.2	97.6	99.7

Table 2 tests this pattern in the low-R^2_τ DGPs across nominal levels. HC1 intervals are close to nominal at 95% and 99% but undercover at 80% in the skewed designs, by up to 4.9 percentage points for IREG with the plug-in and 3.9 for REG, so their 95% coverage is partly a crossing point of a miscalibrated interval. With JK, neither estimator is more than 0.2 points below nominal at any of the three levels in any of these designs; under the heaviest tails it is conservative, with REG+JK covering up to 84.2% at nominal 80%.

4 Discussion

Altogether, our results tackle the comparison of REG and IREG in finite samples through the two most important features one would want from a treatment effect estimator in practice: accuracy of the point estimate, and validity of feasible inference.

Point estimation. We have used second-order expansions to settle the natural question left open by the first-order equivalence of IREG and REG's MSE in a balanced CRT: which estimator is the optimal choice in finite-sample settings, and how significant is the higher-order effect? Theorem 2 responds: there is no universal answer, and the optimality depends on a mix of treatment-effect heterogeneity, covariate shape, residual heteroskedasticity, and nonlinear misspecification. There are important cases where either estimator prevails. In the well-specified Gaussian case, the advantage of IREG grows linearly in p. More generally, however, skewness and heavy tails easily reverse this ordering, particularly when interacted with heteroskedasticity or misspecification. In our scalar lognormal designs, this produces deficiencies of 12, 41, and 166 observations at $s = 0.8, 1, 1.2$, respectively, and at $n = 500$ the simulated MSE of IREG exceeds that of REG by 4.3% and 9.2% in the latter two designs.

These numbers demonstrate that higher-order terms can yield a meaningful difference in practice, particularly with either smaller n or large n and a severely skewed, heavy-tailed, or misspecified DGP. Furthermore, the connection of our work to asymptotic deficiency could be useful for practitioners in quantifying the significance of this effect. For instance, if you estimate Δ for your dataset via prior knowledge or plug-in moments, then the asymptotic cost of using the inferior estimator is approximately $\Delta \cdot c$, where c is the cost per unit, subject to the caveat that follows.

An important caveat here is that, paradoxically, it is the settings where this difference would matter most—heavily skewed or misspecified DGPs, with a large D_p—where the third-order and higher terms can still contribute meaningfully at moderate n (Appendix C). The same higher moments make a plug-in estimate of Δ noisy, and with very heavy tails the moments required by Assumption 1 may not exist. Thus, in those cases our expression is more useful in determining the binary question of optimal estimator and approximate magnitude of difference, rather than as an exact pointer of the difference in MSE that will be observed. Furthermore, while we have quantified this difference in a range of DGPs, and given it the practical interpretation of asymptotic deficiency, our analysis is limited in that we only use synthetic processes. A natural next step is to apply these formulas to real-life datasets using plug-in moments.

Feasible inference. We have examined numerical evidence of feasible inference behavior of both estimators when paired with popular robust standard error estimates HC1 and JK. Importantly, even though IREG is largely the recommended estimator in the statistics literature and HC1 prevails in applied research, this combination is not the most attractive in many practical settings. With low R^2_τ—which we expect to be common in practice—IREG with HC1 and the plug-in correction matches REG+HC1 at the 95% level but is miscalibrated across nominal levels, while REG with JK reaches nominal coverage, up to Monte Carlo error, at every level we examine, with intervals at most 0.5% wider, and up to 14% narrower, than those of IREG with either variance estimator. However, this trend does not hold with large R^2_τ and skewed covariates, in which case REG undercovers with either variance estimator and IREG with JK becomes the most reliable combination, although no procedure reaches nominal coverage in the most extreme of our designs. Our practical recommendation is thus to prefer JK to HC1, and to exercise judgement on the realistic order of R^2_τ when choosing between REG and IREG.

This finding is intuitive; when the linear HTE is only a small portion of the leftover variance from simple regression, interacted regression misattributes and thus miscalibrates it. However, if the linear HTE explains a lot of variance, then adding this term can be genuinely useful. This general idea holds true for both point estimates and feasible inference, although we do not claim a one-to-one relationship between Δ and coverage.

We also acknowledge that this part of our work is based on numerical evidence and not rooted in rigorous theory; a next step in this direction could be computing higher-order expansions for coverage when V^* is replaced by $\hat{V}$.

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A Proofs

Throughout this appendix, indices i, j, k, l, a, b, c, d for Z, T_3, T_4, $\mathcal{E}$ and P refer to coordinates of the whitened covariate vector. We write $G = \|\gamma\|^2$.

Under complete randomization with $n_1 = m$ treated units, write $U_i^{(w)}$ for the vector containing X_i, $Y_i(w)$ and the distinct entries of $X_i X_i^\top$ and $X_i Y_i(w)$, and let $\bar{U}_1$ and $\bar{U}_0$ be the averages of $U_i^{(1)}$ over treated units and of $U_i^{(0)}$ over control units, with population means μ_1 and μ_0.

Lemma 6 (Two-sample representation)**.** *Let $(X_i, Y_i(0), Y_i(1))$ be IID and let the assignment vector W be independent of them, with $\sum_i W_i = m$. Conditionally on W, the treated observations $\{(X_i, Y_i(1)) : W_i = 1\}$ and the control observations $\{(X_i, Y_i(0)) : W_i = 0\}$ are independent IID samples of sizes m and $n - m$ from the laws of $(X, Y(1))$ and $(X, Y(0))$ respectively. Both $\hat{\tau}_{\mathrm{REG}}$ and $\hat{\tau}_{\mathrm{IREG}}$ are of the form $h_{j,r}(\bar{U}_1, \bar{U}_0)$ with $r = m/n$, where $h_{j,r}$ is infinitely differentiable in a neighborhood of (μ_1, μ_0) and its derivatives at (μ_1, μ_0) are rational functions of r without poles in $(0, 1)$.*

Proof. Given W, the units remain IID because W is independent of them; the treated and control index sets are fixed and disjoint, so the two blocks are independent IID samples, observed through $Y_i(1)$ and $Y_i(0)$ respectively. Both estimators are rational functions of per-arm means, second moments and cross-moments: $\hat{\tau}_{\mathrm{IREG}}$ is the difference of the arm-wise intercepts after

centering at $\bar{X} = r\bar{X}_1 + (1-r)\bar{X}_0$, with arm-wise least-squares slopes $\mathcal{S}_w^{-1} s_w$ built from $\bar{U}_w$; $\hat{\tau}_{\mathrm{REG}}$ uses the pooled within-arm slope, whose Gram matrix is $r\mathcal{S}_1 + (1-r)\mathcal{S}_0$. The Gram matrices are invertible at (μ_1, μ_0) because $\Sigma \succ 0$, which gives the smoothness and the dependence on r. □

Lemma 7 (Conditional expansion)**.** *Under Assumption 1, let* $\mathrm{MSE}^{\mathrm{exp}}_{\mathrm{CRT},r}(\hat{\tau}_j)$ *be the second moment of the third-order Taylor polynomial of* $h_{j,r}$ *at* (μ_1, μ_0)*. For every* $\eta \in (0, \frac{1}{2})$*,*

$$\mathrm{MSE}^{\mathrm{exp}}_{\mathrm{CRT},r}(\hat{\tau}_j) = \frac{V_j(r)}{n} + \frac{c_j(r)}{n^2} + O(n^{-3}), \qquad j \in \{\mathrm{REG}, \mathrm{IREG}\}, \tag{13}$$

uniformly in $r = m/n \in [\frac{1}{2} - \eta, \frac{1}{2} + \eta]$*, with* V_j *and* c_j *infinitely differentiable in* r*.*

Proof. The Taylor polynomial is a polynomial of degree three in $(\bar{U}_1 - \mu_1, \bar{U}_0 - \mu_0)$ whose coefficients are smooth in r (Lemma 6); its square has degree six. By Lemma 6 the two blocks are independent IID samples, so each mixed moment of the centered per-arm means is a finite sum of products of joint cumulants of $U^{(1)}$ and $U^{(0)}$ divided by powers of $m = rn$ and $(n - m) = (1 - r)n$. Moments up to order six are finite because the entries of $U^{(w)}$ are products of at most two coordinates of $(X_i, Y_i(w))$ and Assumption 1 bounds their twelfth moments. Collecting terms gives a finite expansion in integer powers of $1/n$ whose coefficients are rational in r without poles on $[\frac{1}{2} - \eta, \frac{1}{2} + \eta]$; the terms beyond n^{-2} are $O(n^{-3})$ uniformly on that interval. □

Lemma 8 (Bernoulli and conditional expansions)**.** *Under Assumption 1 with* $W_i \overset{\mathrm{iid}}{\sim} \mathrm{Bernoulli}(1/2)$ *independent of the units, let* $\bar{W} = n^{-1} \sum_i W_i$*. For every* $\eta \in (0, \frac{1}{2})$*,*

$$\mathrm{MSE}^{\mathrm{exp}}_{\mathrm{Bern}}(\hat{\tau}_j) = \mathbb{E}\Big[\mathrm{MSE}^{\mathrm{exp}}_{\mathrm{CRT},\bar{W}}(\hat{\tau}_j) \mathbf{1}\{|\bar{W} - \tfrac{1}{2}| \le \eta\} \Big] + O(n^{-5/2}).$$

Proof. Let $\mathcal{P}_{\mathrm{Bern}}$ be the Taylor polynomial of Definition 1 under Bernoulli assignment and $\mathcal{P}_{\bar{W}}$ the conditional polynomial of Lemma 7 evaluated at $r = \bar{W}$. Every coordinate of $\bar{\mathcal{V}} - \mu$ is a polynomial of degree at most two in $(\bar{W} - \frac{1}{2}, \bar{U}_1 - \mu_1, \bar{U}_0 - \mu_0)$ and is linear in the per-arm means; for instance,

$$n^{-1} \sum_i W_i U_i^{(1)} - \frac{1}{2}\mu_1 = (\bar{W} - \tfrac{1}{2})\mu_1 + \frac{1}{2}(\bar{U}_1 - \mu_1) + (\bar{W} - \tfrac{1}{2})(\bar{U}_1 - \mu_1).$$

Fix $\eta \in (0, \frac{1}{2})$ and consider the event $|\bar{W} - \frac{1}{2}| \le \eta$. On this event, write

$$\hat{\tau}_j = F_j\big(\bar{W} - \tfrac{1}{2},\, \bar{U}_1 - \mu_1,\, \bar{U}_0 - \mu_0\big)\,,$$

where F_j is smooth. Both $\mathcal{P}_{\mathrm{Bern}}$ and $\mathcal{P}_{\bar{W}}$ agree with the Taylor expansion of F_j at the origin through total degree three: $\mathcal{P}_{\mathrm{Bern}}$ because the map from $(\bar{W}, \bar{U}_1, \bar{U}_0)$ to $\bar{\mathcal{V}}$ is polynomial, and $\mathcal{P}_{\bar{W}}$ because it is the conditional expansion with $r = \bar{W}$. Hence their difference consists only of terms of total degree at least four in quantities that are $O_{L^k}(n^{-1/2})$. Under Assumption 1,

$$\|\mathcal{P}_{\mathrm{Bern}} - \mathcal{P}_{\bar{W}}\|_{L^2} = O(n^{-2}).$$

Also,

$$\|\mathcal{P}_{\mathrm{Bern}} + \mathcal{P}_{\bar{W}} - 2\tau\|_{L^2} = O(n^{-1/2}),$$

so Cauchy–Schwarz gives

$$\big|\mathbb{E}\left[(\mathcal{P}_{\mathrm{Bern}} - \tau)^2 - (\mathcal{P}_{\bar{W}} - \tau)^2\right] \mathbf{1}\{|\bar{W} - \tfrac{1}{2}| \le \eta\}\big| = O(n^{-5/2}).$$

Finally, Hoeffding's inequality gives

$$\Pr\left(|\bar{W} - \tfrac{1}{2}| > \eta\right) \le 2e^{-2n\eta^2},$$

so the complement contributes an exponentially small term, which is $O(n^{-5/2})$. This proves the claim. □

The proof of Theorem 2 combines these lemmas, a reduction from the CRT to the Bernoulli design, and the Bernoulli calculation.

Theorem 9. *Suppose the Bernoulli design has second-order difference*

$$\mathrm{MSE}^{\exp}_{\mathrm{Bern}}(\hat{\tau}_{\mathrm{IREG}}) - \mathrm{MSE}^{\exp}_{\mathrm{Bern}}(\hat{\tau}_{\mathrm{REG}}) = \frac{D_{\mathrm{Bern}}}{n^2} + O(n^{-5/2}) \tag{14}$$

at $\pi = 1/2$, and let Assumption 1 hold. Then, under complete randomization with $n_1 = \lfloor n/2 \rfloor$ treated units,

$$\mathrm{MSE}^{\exp}_{\mathrm{CRT}}(\hat{\tau}_{\mathrm{IREG}}) - \mathrm{MSE}^{\exp}_{\mathrm{CRT}}(\hat{\tau}_{\mathrm{REG}}) = \frac{D_{\mathrm{Bern}} + 4G}{n^2} + O(n^{-3}). \tag{15}$$

Proof. Let $n_1 = \sum_{i=1}^n W_i$ under the Bernoulli design and write $r = n_1/n$. Conditional on $n_1 = m$, the treatment assignment has the same distribution as a CRT with treatment fraction r. Fix $\eta \in (0, \frac{1}{2})$. By Lemma 8, $\mathrm{MSE}^{\exp}_{\mathrm{Bern}}$ is, up to $O(n^{-5/2})$, the average over n_1 of $\mathrm{MSE}^{\exp}_{\mathrm{CRT},r}$ restricted to $|r - \frac{1}{2}| \le \eta$, and by Lemma 7 the latter admits the expansion (13) uniformly on that set, with V_j and c_j infinitely differentiable. At treatment fraction r the first-order variance difference is, by Proposition 1,

$$V_{\mathrm{IREG}}(r) - V_{\mathrm{REG}}(r) = -\frac{(1-2r)^2}{r(1-r)}G,$$

and

$$\frac{(1-2r)^2}{r(1-r)} = 16u^2 + O(u^4), \qquad u = r - \tfrac{1}{2}.$$

Under Bernoulli randomization with $\pi = 1/2$, $\mathbb{E}[u^2] = 1/(4n)$ and $\mathbb{E}[u^4] = O(n^{-2})$, so

$$\frac{1}{n}\mathbb{E}\big[V_{\mathrm{IREG}}(r) - V_{\mathrm{REG}}(r)\big] = -\frac{4G}{n^2} + O(n^{-3}).$$

For the second-order coefficients, a second-order Taylor expansion of $c_{\mathrm{IREG}} - c_{\mathrm{REG}}$ around $1/2$ together with $\mathbb{E}[u] = 0$ and $\mathbb{E}[u^2] = O(n^{-1})$ gives

$$\mathbb{E}[c_{\mathrm{IREG}}(r) - c_{\mathrm{REG}}(r)] = c_{\mathrm{IREG}}(1/2) - c_{\mathrm{REG}}(1/2) + O(n^{-1}).$$

Averaging (13) over n_1 therefore yields

$$\mathrm{MSE}^{\exp}_{\mathrm{Bern}}(\hat{\tau}_{\mathrm{IREG}}) - \mathrm{MSE}^{\exp}_{\mathrm{Bern}}(\hat{\tau}_{\mathrm{REG}}) = \frac{c_{\mathrm{IREG}}(1/2) - c_{\mathrm{REG}}(1/2) - 4G}{n^2} + O(n^{-5/2}),$$

so

$$D_{\mathrm{Bern}} = c_{\mathrm{IREG}}(1/2) - c_{\mathrm{REG}}(1/2) - 4G.$$

Under balanced complete randomization, $r = 1/2$ for even n, giving

$$D_{\mathrm{CRT}} = c_{\mathrm{IREG}}(1/2) - c_{\mathrm{REG}}(1/2) = D_{\mathrm{Bern}} + 4G.$$

For even n, Lemma 7 therefore gives directly

$$\mathrm{MSE}^{\exp}_{\mathrm{CRT}}(\hat{\tau}_{\mathrm{IREG}}) - \mathrm{MSE}^{\exp}_{\mathrm{CRT}}(\hat{\tau}_{\mathrm{REG}}) = \frac{D_{\mathrm{Bern}} + 4G}{n^2} + O(n^{-3}).$$

For odd n, $r - 1/2 = O(n^{-1})$. The first-order variance difference is quadratic in $r - 1/2$, so its contribution to the MSE is $O(n^{-3})$; the smoothness of $c_{\mathrm{IREG}} - c_{\mathrm{REG}}$ gives an $O(n^{-1})$ change in the second-order coefficient, which is also $O(n^{-3})$ after division by n^2. Thus the same expansion holds for odd n. □

Theorem 10. *Let Assumption 1 hold, except that* $W_i \overset{\text{iid}}{\sim}$ *Bernoulli*$(1/2)$ *independently of* $(X_i, Y_i(0), Y_i(1))$. *Define*

$$T_{3,ijk} = \mathbb{E}[Z_i Z_j Z_k], \quad T_{4,ijkl} = \mathbb{E}[Z_i Z_j Z_k Z_l], \quad q_i = T_{3,ijk}\delta_{jk}, \quad L_{ij} = T_{3,iab}T_{3,jab}, \tag{16}$$

write $S = q = \mathbb{E}[Z\|Z\|^2]$, *and set* $\mathcal{E} = \mathbb{E}[ZZ^\top\varepsilon]$ *and* $P = \mathbb{E}[ZZ^\top(W - \frac{1}{2})\varepsilon]$. *For the scalar contractions*

$$G = \|\gamma\|^2, \qquad F = \gamma_i\gamma_j T_{4,ijkk}, \qquad Q = (\gamma^\top q)^2, \qquad \mathcal{R} = \gamma^\top L\gamma, \tag{17}$$

$$A_P = \gamma_i T_{3,ijk} P_{jk}, \qquad B_P = (\gamma^\top q)\,\mathrm{tr}(P), \qquad C_P = \gamma^\top P q, \tag{18}$$

we have

$$\mathrm{MSE}^{\exp}_{\mathrm{Bern}}(\hat{\tau}_{\mathrm{IREG}}) - \mathrm{MSE}^{\exp}_{\mathrm{Bern}}(\hat{\tau}_{\mathrm{REG}}) = \frac{D_{\mathrm{Bern}}}{n^2} + O(n^{-5/2}), \tag{19}$$

where

$$D_{\mathrm{Bern}} = -(2p+7)G + F - Q - 3\mathcal{R} - 16A_P - 8B_P + 8C_P + 8\,\mathrm{tr}(\mathcal{E}^2) + 8S^\top\mathbb{E}[Z\varepsilon^2]. \tag{20}$$

Proof. Both REG and IREG contain the regressors 1, W and X. Subtracting any linear combination of these regressors from the outcome changes the fitted coefficient on W by the corresponding amount while leaving the estimation error unchanged, so we may subtract the population main effects and the term τW_i and work with the exact representation

$$Y_i = W_i X_i^\top\{\beta^*_{(1)} - \beta^*_{(0)}\} + \varepsilon_i.$$

Whitening as in (3) gives the canonical representation $Y_i = W_i\gamma^\top Z_i + \varepsilon_i$, and we write

$$W_i = \tfrac{1}{2} + \omega_i, \qquad \omega_i \in \{-\tfrac{1}{2}, \tfrac{1}{2}\}, \qquad \mathbb{E}[\omega_i] = 0, \qquad \omega_i^2 = \tfrac{1}{4}.$$

Let $\mathcal{V}_i$ denote the vector of raw-moment coordinates needed to evaluate the REG or IREG coefficient map, and adopt the notation of Definition 1, so that $\hat{\tau} = g(\bar{\mathcal{V}})$ and $\tilde{\tau}_n$ is given by (8). Let

$$\Sigma^{\mathcal{V}}_{ij} = \mathrm{Cov}(\mathcal{V}_i, \mathcal{V}_j), \qquad \mu^{\mathcal{V}}_{3,ijk} = \mathbb{E}[(\mathcal{V}_i - \mu_i)(\mathcal{V}_j - \mu_j)(\mathcal{V}_k - \mu_k)]$$

be the covariance and third central-moment tensors of the moment vector. Expanding the square of (8) and using the second- and third-order moments of sample means gives the order-n^{-1} correction to $n\,\mathrm{MSE}^{\exp}$ as $\mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3$ with

$$\mathcal{T}_1 = a_i H_{jk}\mu^{\mathcal{V}}_{3,ijk}, \tag{21}$$

$$\mathcal{T}_2 = \frac{1}{4}H_{ij}H_{kl}\left(\Sigma^{\mathcal{V}}_{ik}\Sigma^{\mathcal{V}}_{jl} + \Sigma^{\mathcal{V}}_{il}\Sigma^{\mathcal{V}}_{jk} + \Sigma^{\mathcal{V}}_{ij}\Sigma^{\mathcal{V}}_{kl}\right), \tag{22}$$

$$\mathcal{T}_3 = \frac{1}{3}a_i A_{3,jkl}\left(\Sigma^{\mathcal{V}}_{ij}\Sigma^{\mathcal{V}}_{kl} + \Sigma^{\mathcal{V}}_{ik}\Sigma^{\mathcal{V}}_{jl} + \Sigma^{\mathcal{V}}_{il}\Sigma^{\mathcal{V}}_{jk}\right), \tag{23}$$

the remaining terms being $O(n^{-3/2})$ in $n\,\mathrm{MSE}^{\mathrm{exp}}$ under Assumption 1. The difference between the order-n^{-2} MSE coefficients of REG and IREG is obtained by evaluating these three contractions for the two estimators.

The population Gram matrices and their inverses in the whitened coordinates are

$$A_R = \begin{pmatrix} 1 & 0 & 1/2 \\ 0 & I_p & 0 \\ 1/2 & 0 & 1/2 \end{pmatrix}, \qquad A_R^{-1} = \begin{pmatrix} 2 & 0 & -2 \\ 0 & I_p & 0 \\ -2 & 0 & 4 \end{pmatrix}, \qquad A_I = \begin{pmatrix} 1 & 0 & 1/2 & 0 \\ 0 & I_p & 0 & I_p/2 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & I_p/2 & 0 & I_p/2 \end{pmatrix},$$

$$A_I^{-1} = \begin{pmatrix} 2 & 0 & -2 & 0 \\ 0 & 2I_p & 0 & -2I_p \\ -2 & 0 & 4 & 0 \\ 0 & -2I_p & 0 & 4I_p \end{pmatrix}, \qquad \theta_R = (0, \gamma/2, 0), \qquad \theta_I = (0, 0, 0, \gamma).$$

Writing $\Omega = A^{-1}$ and $\theta = \Omega b$, differentiation of $A\theta = b$ gives the exact derivative identities

$$D\theta[h] = \Omega(db - dA\,\theta),$$

$$D^2\theta[h_1, h_2] = -\Omega dA_1 \Omega(db_2 - dA_2\theta) - \Omega dA_2 \Omega(db_1 - dA_1\theta),$$

$$D^3\theta[h_1, h_2, h_3] = \sum_{\varsigma \in \mathfrak{S}_3} \Omega dA_{\varsigma_1} \Omega dA_{\varsigma_2} \Omega(db_{\varsigma_3} - dA_{\varsigma_3}\theta).$$

REG's estimator is the treatment coefficient, $g_{\mathrm{REG}}(v) = \theta_W(v)$. IREG's is $g_{\mathrm{IREG}}(v) = \theta_W(v) + \bar{z}(v)^\top \theta_{WZ}(v)$, where $\bar{z}(v)$ is the full-sample covariate mean, a linear coordinate of v that vanishes at μ in whitened coordinates, and $\theta_{WZ} = \gamma$ at μ. The product rule gives

$$a_{\mathrm{IREG}}[h] = D\theta_W[h] + D\bar{z}[h]^\top \gamma,$$

$$H_{\mathrm{IREG}}[h_1, h_2] = D^2\theta_W[h_1, h_2] + D\bar{z}[h_1]^\top D\theta_{WZ}[h_2] + D\bar{z}[h_2]^\top D\theta_{WZ}[h_1],$$

$$A_{3,\mathrm{IREG}}[h_1, h_2, h_3] = D^3\theta_W[h_1, h_2, h_3] + \sum_{\{i,j,k\}=\{1,2,3\}} D\bar{z}[h_i]^\top D^2\theta_{WZ}[h_j, h_k],$$

the sum running over the three choices of i. The covariance and third-moment blocks of $\mathcal{V}_i$ are products of moments of $(Z_i, \omega_i, \varepsilon_i)$ under the canonical representation of Section 2, with W_i marginally Bernoulli(1/2).

Substituting these derivative tensors and the covariance and third-moment blocks of $\mathcal{V}_i$ into (21)–(23), we reduce the resulting indexed expressions using tensor symmetries, dummy-index relabeling, and the identities implied by $\mathbb{E}[ZZ^\top] = I_p$ and $\omega_i^2 = 1/4$. This gives

$$\Delta\mathcal{T}_1 = 2G + 2F + 8\Xi_\gamma, \tag{24}$$

$$\Delta\mathcal{T}_2 = -3G - F - Q - \mathcal{R} - 8\Xi_\gamma - 8A_P - 8B_P, \tag{25}$$

$$\Delta\mathcal{T}_3 = -(2p+6)G - 2\mathcal{R} - 8A_P + 8C_P + 8\,\mathrm{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2], \tag{26}$$

where

$$\Xi_\gamma = \mathbb{E}[(\gamma^\top Z)\|Z\|^2 \omega\varepsilon]$$

with $\omega = W - \frac{1}{2}$. The accompanying verification script enumerates the full Cartesian products of the expansion directions and reduces the resulting indexed expressions using only tensor symmetries, dummy-index relabeling, and the identities implied by $\mathbb{E}[ZZ^\top] = I_p$ and $\omega_i^2 = 1/4$, with the dimension p kept symbolic. The symbolic remainder in each of the three rows is identically

zero; the verification is implemented in `verification/general_p_symbolic_certificate.py` in the replication repository and runs in under a minute.

Summing the three rows, the Ξ_γ terms cancel and (20) follows.

The division of labour is as follows. Derived by hand and displayed above: the expansion terms (21)–(23), the Gram matrices and coefficients at μ, the derivative identities, the product-rule structure of the IREG centering functional, and the reduction rules. Verified symbolically by the accompanying implementation: the substitution of these objects into (21)–(23) and the reduction of the resulting indexed sums to the rows (24)–(26), with symbolic p and zero remainder. The resulting constant is checked independently by Monte Carlo in Appendix B. □

Theorems 9 and 10 give

$$D_p = -(2p+3)G + F - Q - 3\mathcal{R} - 16A_P - 8B_P + 8C_P + 8\,\mathrm{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2], \tag{27}$$

which Lemma 11 rewrites in the matrix form of Theorem 2.

Lemma 11 (Equivalent matrix form)**.** *With K, S, S_γ, $\mathcal{E}$ and $\mathcal{E}_W$ as in Theorem 2,*

$$G = \|\gamma\|^2, \qquad F = \gamma^\top K\gamma, \qquad Q = (\gamma^\top S)^2, \qquad \mathcal{R} = \|S_\gamma\|_F^2,$$
$$A_P = \mathrm{tr}(S_\gamma \mathcal{E}_W), \qquad B_P = (\gamma^\top S)\,\mathrm{tr}(\mathcal{E}_W), \qquad C_P = \gamma^\top \mathcal{E}_W S.$$

Proof. First,

$$F = \gamma_i\gamma_j T_{4,ijkl}\delta_{kl} = \gamma^\top \mathbb{E}[ZZ^\top \|Z\|^2]\gamma = \gamma^\top K\gamma,$$

and

$$S = \mathbb{E}[Z_i Z_j Z_j]_{i=1}^p = q$$

gives

$$Q = (\gamma^\top q)^2 = (\gamma^\top S)^2.$$

Next,

$$(S_\gamma)_{ab} = \mathbb{E}[Z_a Z_b (Z^\top \gamma)] = T_{3,abk}\gamma_k,$$

so

$$\|S_\gamma\|_F^2 = \sum_{a,b} (S_\gamma)_{ab}^2 = \gamma_k\gamma_\ell T_{3,abk}T_{3,ab\ell} = \gamma_k\gamma_\ell L_{k\ell} = \mathcal{R}.$$

Since S_γ and $\mathcal{E}_W$ are symmetric,

$$\mathrm{tr}(S_\gamma \mathcal{E}_W) = (S_\gamma)_{jk}(\mathcal{E}_W)_{jk} = \gamma_i T_{3,ijk} P_{jk} = A_P;$$

finally,

$$B_P = (\gamma^\top q)\,\mathrm{tr}(P) = (\gamma^\top S)\,\mathrm{tr}(\mathcal{E}_W)$$

and

$$C_P = \gamma^\top P q = \gamma^\top \mathcal{E}_W S.$$

□

We close with an exact identity that is used by the variance-reduced verification of Appendix B.

Lemma 12 (Exact difference identity)**.** *Let* $\bar{X}_w$ *and*

$$\mathcal{S}_w = \sum_{i:W_i=w} (X_i - \bar{X}_w)(X_i - \bar{X}_w)^\top$$

denote the within-arm means and scatter matrices, let

$$\hat{\beta}_w = \mathcal{S}_w^{-1} \sum_{i:W_i=w} (X_i - \bar{X}_w)(Y_i - \bar{Y}_w),$$

and write $d_X = \bar{X}_1 - \bar{X}_0$*. Then, whenever* $\mathcal{S}_0$*,* $\mathcal{S}_1$ *and* $\mathcal{S}_0 + \mathcal{S}_1$ *are invertible and* $n_1 = n_0$*,*

$$\hat{\tau}_{\mathrm{REG}} = \bar{Y}_1 - \bar{Y}_0 - d_X^\top (\mathcal{S}_1 + \mathcal{S}_0)^{-1}(\mathcal{S}_1\hat{\beta}_1 + \mathcal{S}_0\hat{\beta}_0),$$

$$\hat{\tau}_{\mathrm{IREG}} = \bar{Y}_1 - \bar{Y}_0 - d_X^\top \frac{\hat{\beta}_1 + \hat{\beta}_0}{2},$$

and consequently

$$\hat{\tau}_{\mathrm{IREG}} - \hat{\tau}_{\mathrm{REG}} = \tfrac{1}{2}\, d_X^\top (\mathcal{S}_1 + \mathcal{S}_0)^{-1}(\mathcal{S}_1 - \mathcal{S}_0)(\hat{\beta}_1 - \hat{\beta}_0) = O_P(n^{-1}).$$

Proof. Both expressions for the estimators are the Frisch–Waugh–Lovell representations of (1): the coefficient on X in the REG regression is the pooled within-arm slope $(\mathcal{S}_1 + \mathcal{S}_0)^{-1}(\mathcal{S}_1\hat{\beta}_1 + \mathcal{S}_0\hat{\beta}_0)$, and IREG fits the two arms separately, so with $n_1 = n_0$ the overall covariate mean satisfies

$$\bar{X}_1 - \bar{X} = d_X/2 = -(\bar{X}_0 - \bar{X}).$$

Subtracting,

$$\hat{\tau}_{\mathrm{IREG}} - \hat{\tau}_{\mathrm{REG}} = d_X^\top \left[(\mathcal{S}_1 + \mathcal{S}_0)^{-1}(\mathcal{S}_1\hat{\beta}_1 + \mathcal{S}_0\hat{\beta}_0) - \tfrac{1}{2}(\hat{\beta}_1 + \hat{\beta}_0)\right]$$

$$= \tfrac{1}{2} d_X^\top (\mathcal{S}_1 + \mathcal{S}_0)^{-1}(\mathcal{S}_1 - \mathcal{S}_0)(\hat{\beta}_1 - \hat{\beta}_0).$$

The stochastic order follows from

$$d_X = O_P(n^{-1/2}), \qquad (\mathcal{S}_1 + \mathcal{S}_0)^{-1}(\mathcal{S}_1 - \mathcal{S}_0) = O_P(n^{-1/2}), \qquad \hat{\beta}_1 - \hat{\beta}_0 = O_P(1).$$

□

In whitened coordinates the leading term of this difference is

$$\tfrac{1}{4} d_Z^\top (A_1 - A_0)\gamma,$$

with A_w the arm-w mean of $ZZ^\top - I_p$; this is the control variate used in (28) below.

Lemma 13 (Leading expansion bias)**.** *Under Assumption 1 and balanced complete randomization,*

$$\mathbb{E}[\tilde{\tau}_{n,\mathrm{IREG}} - \tau] = -\frac{4\,\mathrm{tr}(\mathcal{E}_W)}{n} + O(n^{-2}),$$

$$\mathbb{E}[\tilde{\tau}_{n,\mathrm{REG}} - \tau] = -\frac{4\,\mathrm{tr}(\mathcal{E}_W) + \gamma^\top S}{n} + O(n^{-2}).$$

Proof. Work in whitened coordinates with $m = n/2$ units per arm, arm slopes $b_1 - b_0 = \gamma$, projection residuals $\varepsilon_i(w)$ and within-arm scatter matrices $\mathcal{S}_w$. By Remark 3, the expansion bias is the mean of the quadratic part of $\hat{\tau} - \tau$, up to $O(n^{-2})$. For IREG,

$$\bar{Z}_1 - \bar{Z} = -(\bar{Z}_0 - \bar{Z}) = \frac{1}{2}(\bar{Z}_1 - \bar{Z}_0),$$

so

$$\hat{\tau}_{\mathrm{IREG}} - \tau = \bar{\varepsilon}_1 - \bar{\varepsilon}_0 + \gamma^\top \bar{Z} - \tfrac{1}{2}(\bar{Z}_1 - \bar{Z}_0)^\top\{(\hat{b}_1 - b_1) + (\hat{b}_0 - b_0)\},$$

and the linear part of $\hat{b}_w - b_w$ is

$$m^{-1} \sum_{i:W_i=w} Z_i \varepsilon_i(w),$$

which enters the quadratic part of $\hat{\tau}_{\mathrm{IREG}} - \tau$ through its product with $\bar{Z}_1 - \bar{Z}_0$. The arms are independent (Lemma 6), so

$$\mathbb{E}[(\bar{Z}_1 - \bar{Z}_0)^\top(\hat{b}_1 - b_1)] = m^{-1}\mathbb{E}[\|Z\|^2\varepsilon(1)] + O(n^{-2})$$

and

$$\mathbb{E}[(\bar{Z}_1 - \bar{Z}_0)^\top(\hat{b}_0 - b_0)] = -m^{-1}\mathbb{E}[\|Z\|^2\varepsilon(0)] + O(n^{-2}),$$

which gives

$$-(2m)^{-1}\operatorname{tr}\{\mathbb{E}[ZZ^\top\varepsilon(1)] - \mathbb{E}[ZZ^\top\varepsilon(0)]\} = -\frac{4\operatorname{tr}(\mathcal{E}_W)}{n}.$$

For REG, the pooled slope $\hat{b}$ estimates $\bar{b} = (b_1 + b_0)/2$ and

$$\hat{\tau}_{\mathrm{REG}} - \tau = \bar{\varepsilon}_1 - \bar{\varepsilon}_0 + \tfrac{1}{2}\gamma^\top(\bar{Z}_1 + \bar{Z}_0) - (\bar{Z}_1 - \bar{Z}_0)^\top(\hat{b} - \bar{b}).$$

The pooled residual of unit i is $\varepsilon_i(W_i) + (W_i - \frac{1}{2})\gamma^\top Z_i$ up to centering, so the linear part of $\hat{b} - \bar{b}$ is

$$n^{-1}\sum_i Z_i\{\varepsilon_i(W_i) + (W_i - \tfrac{1}{2})\gamma^\top Z_i\},$$

which again enters the quadratic part of $\hat{\tau}_{\mathrm{REG}} - \tau$ through its product with $\bar{Z}_1 - \bar{Z}_0$, and

$$\mathbb{E}[(\bar{Z}_1 - \bar{Z}_0)^\top(\hat{b} - \bar{b})] = n^{-1}\left\{\operatorname{tr}\mathbb{E}[ZZ^\top\varepsilon(1)] - \operatorname{tr}\mathbb{E}[ZZ^\top\varepsilon(0)] + \gamma^\top S\right\} + O(n^{-2}),$$

which gives the second display. □

B Numerical verification of Theorem 2

The symbolic certificate verifies the algebra of Theorem 10 exactly. The role of simulation is different: it checks that the constant D_p governs the sampling behavior of the estimators, that the CRT correction in Theorem 9 is correct, and that no terms were lost in the derivation. Simulation cannot prove the theorem, so we use several checks that are designed to distinguish the stated formula from plausible alternatives.

Exact targets. We use data-generating processes with whitened covariates having independent coordinates with known moments. The oracle residuals take the form

$$\varepsilon(w) = g_w(Z) + s_w(Z)\eta,$$

where η is independent of Z with mean zero and unit variance, g_w is a polynomial orthogonal to $(1, Z)$, and s_w^2 is a polynomial. Each term in D_p is therefore a finite combination of population moments and can be computed in exact rational arithmetic. The code also checks $\mathbb{E}[g_w] = \mathbb{E}[Zg_w] = 0$, so that violations of the orthogonality conditions cannot pass unnoticed.

We use six designs, chosen so that every nontrivial block of D_p is active in at least one design: Gaussian homoskedastic covariates; centered-exponential covariates with heterogeneity; centered-exponential covariates with heteroskedasticity and $\gamma = 0$; centered-exponential covariates with heterogeneity, arm-specific misspecification, and arm-specific heteroskedasticity; Gaussian covariates with arm-specific misspecification and heteroskedasticity; and a mixed skewed/Gaussian design. These designs are analytically tractable and differ from the inference designs of Section 3; their exact definitions are in `simulations/dgps.py` of the replication repository.

A variance-reduced estimator of D_p. The natural statistic $n^2\{\widehat{\mathrm{MSE}}_{\mathrm{IREG}} - \widehat{\mathrm{MSE}}_{\mathrm{REG}}\}$ has an unfavorable variance rate. By Lemma 12, the leading term of $\hat{\tau}_{\mathrm{IREG}} - \hat{\tau}_{\mathrm{REG}}$ is

$$d_1 = \tfrac{1}{4} d_Z^\top (A_1 - A_0)\gamma,$$

which is $O_P(n^{-1})$, while the common first-order error is

$$L = \gamma^\top \bar{Z} + \bar{\varepsilon}_1 - \bar{\varepsilon}_0 = O_P(n^{-1/2}).$$

Thus

$$e_{\mathrm{IREG}}^2 - e_{\mathrm{REG}}^2 = 2d_1 L + O_P(n^{-2}),$$

so the leading term of the scaled statistic is $2n^2 d_1 L = O_P(\sqrt{n})$. Under Assumption 1, this leading term has variance $O(n)$, so the corresponding per-replication standard deviation is $O(\sqrt{n})$; in the Gaussian design below, it increases with n as expected.

We remove this leading fluctuation with a control variate whose finite-n expectation is known exactly. Let $d_Z = \bar{Z}_1 - \bar{Z}_0$, let A_w be the arm-w mean of $ZZ^\top - I_p$, and define

$$d_1 = \tfrac{1}{4} d_Z^\top (A_1 - A_0)\gamma, \qquad L = \gamma^\top \bar{Z} + \bar{\varepsilon}_1 - \bar{\varepsilon}_0.$$

For an arm of size $m = n/2$, the expectation of a product of three centered IID sample means is the corresponding third joint cumulant divided by m^2, giving the exact identity

$$\mathbb{E}[d_1 L] = \frac{1}{n^2}\,\mathbb{E}\Big[(\|Z\|^2 - 1)(\gamma^\top Z)\{\gamma^\top Z + \varepsilon(1) - \varepsilon(0)\}\Big]. \tag{28}$$

Hence

$$\hat{D}_p = \frac{1}{B}\sum_{b=1}^{B}\Big[n^2\big(e_{\mathrm{IREG},b}^2 - e_{\mathrm{REG},b}^2\big) - 2n^2 d_{1,b} L_b\Big] + 2n^2\,\mathbb{E}[d_1 L]$$

is unbiased for the same finite-n scaled MSE difference as the raw statistic. Moreover,

$$n^2\{e_{\mathrm{IREG}}^2 - e_{\mathrm{REG}}^2 - 2d_1 L\} = O_P(1),$$

so its per-replication variance is $O(1)$ rather than $O(n)$.

Both statistics are reported by the replication code. In the Gaussian design, the raw per-replication standard deviation increases from 151 at $n = 200$ to 293 at $n = 800$, while the control-variate standard deviation is 56 and 55, respectively. At $n = 3200$, the control-variate Monte Carlo standard error is ten times smaller than that of the raw statistic.

Results. Table 3 reports the exact targets and the control-variate estimates at $n \in \{200, 800, 3200\}$, using $B = 400{,}000$ replications for the two smaller sample sizes and $B = 120{,}000$ for $n = 3200$. The control variate keeps the per-replication standard deviation bounded, making the Monte Carlo errors small enough to resolve the finite-n remainder.

The largest discrepancy occurs in the all-blocks design (fourth row). There the estimate is 32.40 at $n = 200$, compared with the target 44.52, a difference of 12.1 (about 22 standard errors). The difference falls to 5.7 at $n = 800$ and 2.55 at $n = 3200$. The Bernoulli constant $D_{\mathrm{Bern}} = D_p - 4G = 41.32$ is within one standard error of the $n = 3200$ estimate, so that sample size alone does not distinguish D_{Bern} from D_p in this design. The successive increments, 6.4 and 3.1, shrink by a factor of about two as n quadruples. An $O(n^{-3})$ remainder in the expansion MSE alone would shrink them by a factor of four, so in this design the finite-sample MSE apparently also differs from the expansion MSE at a slower rate; we do not extrapolate. The ablation in Table 4 separates D_p from D_{Bern} more clearly in other designs.

At $n = 3200$, all six estimates are within 2.6 standard errors of their targets. The two largest deviations are $z = -2.6$ in the all-blocks design and $z = +2.3$ in the $\gamma = 0$ heteroskedastic design. In the latter, the estimate moves from 29.82 at $n = 200$ and 31.73 at $n = 800$ to 33.25 at $n = 3200$. With only six designs, deviations of this size are compatible with Monte Carlo error. The results are consistent with a remainder that is smaller than the n^{-2} leading term, but three sample sizes are not enough to estimate its rate.

Table 3: Verification of Theorem 2. Targets are exact; estimates are the control-variate estimator with Monte Carlo standard errors in parentheses. $B = 400{,}000$ for $n \in \{200, 800\}$ and $B = 120{,}000$ for $n = 3200$. Parentheses after a design name list the blocks of D_p that are active, in the notation of Theorem 10 and Lemma 11 ($F = \gamma^\top K \gamma$, $Q = (\gamma^\top S)^2$, $\mathcal{R} = \|S_\gamma\|_F^2$, $E = \mathcal{E}$, $E_W = \mathcal{E}_W$); the mixed design has heterogeneity along its skewed coordinate and heteroskedastic residuals, so the covariate-shape blocks and $8S^\top \mathbb{E}[Z\varepsilon^2]$ are active.

DGP (active blocks)	D_p (exact)	$n = 200$	$n = 800$	$n = 3200$
Gaussian, homoskedastic, $p = 3$ (F)	-5.00	-4.70 (0.09)	-4.99 (0.09)	-5.17 (0.16)
Exponential covariates, homoskedastic (F, Q, R)	-20.25	-17.17 (0.21)	-19.60 (0.22)	-19.75 (0.39)
Exponential, heteroskedastic, $\gamma = 0$	32.00	29.82 (0.28)	31.73 (0.29)	33.25 (0.54)
Exponential, HTE + arm-specific misspec. (all)	44.52	32.40 (0.56)	38.82 (0.55)	41.97 (0.98)
Gaussian, arm-specific misspec. ($\mathrm{tr}\, E^2, E_W$)	-2.32	-2.36 (0.09)	-2.34 (0.09)	-2.45 (0.16)
Mixed skewed/Gaussian, heteroskedastic	28.00	25.94 (0.27)	27.85 (0.28)	28.27 (0.53)

Power: what the verification would reject. A useful verification should be able to distinguish the stated formula from plausible alternatives. Table 4 reports z-statistics at $n = 3200$ for the exact D_p and several perturbed formulas: the Bernoulli constant $D_{\mathrm{Bern}} = D_p - 4G$, which omits the CRT correction; the Gaussian special case $-(p+1)G$; and versions of D_p with individual blocks removed.

The exact formula has $|z| \le 2.6$ in every design, whereas each alternative has $|z| > 3$ in at least one design, in some cases by tens of standard errors. The check has limited power when an alternative differs from D_p by less than the residual finite-n bias. In the all-blocks design, for example, $D_{\mathrm{Bern}} = D_p - 3.2$ is not separated from D_p at $n = 3200$, which is why multiple designs are useful.

Without the control variate, the standard errors at $n = 3200$ would be 1.5 to 3.8, four to ten times larger in the heterogeneous designs. The Bernoulli constant would then not be separated

from D_p in any design ($|z| \leq 3$ throughout), so the simulation would have little power to detect an omitted CRT correction.

Table 4: Ablation: $z = (\hat{D}_p - D^{\text{alt}})/\widehat{\text{SE}}$ at $n = 3200$ for each design and candidate constant. Values with $|z| > 3$ indicate rejection of that candidate. Each "drop" column removes one block from D_p: the term $-3\mathcal{R}$, the term $-Q$, the residual terms $8\,\text{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2]$ ("resid."), or the three terms involving $\mathcal{E}_W$.

DGP	D_p	$D_p - 4G$	$-(p+1)G$	drop $-3\mathcal{R}$	drop $-Q$	drop resid.	drop $\mathcal{E}_W$
Gaussian, homoskedastic, $p = 3$ (F)	-1.0	$+29.8$	-1.0	-1.0	-1.0	-1.0	-1.0
Exponential covariates, homoskedastic (F, Q, R)	$+1.3$	$+14.2$	-41.3	-37.4	-21.9	$+1.3$	$+1.3$
Exponential, heteroskedastic, $\gamma = 0$	$+2.3$	$+2.3$	$+61.1$	$+2.3$	$+2.3$	$+61.1$	$+2.3$
Exponential, HTE + arm-specific misspec. (all)	-2.6	$+0.7$	$+45.1$	-12.4	-3.2	$+65.8$	-17.7
Gaussian, arm-specific misspec. ($\text{tr}\,E^2, E_W$)	-0.8	$+23.5$	$+3.4$	-0.8	-0.8	$+3.4$	-0.8
Mixed skewed/Gaussian, heteroskedastic	$+0.5$	$+4.2$	$+55.7$	-5.1	-1.4	$+60.3$	$+0.5$

Further covariate families and dimensions. The six designs above have $p \leq 3$ and independent coordinates. We also consider four covariate families at $p \in \{2, 5, 8\}$, including dependent coordinates (Table 5).

For Gaussian covariates, $D_p = -(p+1)G$. For Rademacher covariates, $K = pI_p$ and $D_p = -(p+3)G$. For a symmetric scale mixture of normals, where 2% of units have six times the standard deviation,

$$K = (p+2)\tilde{\kappa} I_p, \qquad \tilde{\kappa} = \frac{\mathbb{E}[s^4]}{\mathbb{E}[s^2]^2},$$

and

$$D_p = G\{(p+2)\tilde{\kappa} - (2p+3)\}.$$

For Gaussian covariates with quadratic misspecification $c(X^\top X - \text{tr}\,\Sigma)$, where $c = 0.3$ and $\Sigma_{jk} = 0.3^{|j-k|}$,

$$\mathcal{E} = 2c\Sigma, \qquad D_p = -(p+1)G + 32c^2\,\text{tr}(\Sigma^2).$$

Every family uses homoskedastic $N(0,1)$ noise shared by the potential outcomes, with $G = 0.49$ (1.96 for the Rademacher family). Because both estimators are invariant to invertible linear transformations of the covariates, we simulate whitened covariates; dependence enters through γ and, for the misspecified family, through $\text{tr}(\Sigma^2)$.

The targets are exact. Recomputing them from the matrix form of Theorem 2 using 10^6 Monte Carlo draws reproduces them to within 5%, with the largest discrepancy in the scale mixture, where fourth moments are harder to estimate. Using exact targets avoids this additional Monte Carlo error.

With $n = 4{,}000$ and $B = 20{,}000$ replications per design, every estimate is within 1.55 standard errors of its target, and the control variate reduces the standard errors by factors of 3.9 to 11.4 relative to the raw estimator on the same replications.

Figure 1 shows the z-statistics for all 18 checks: the designs of Table 3 at $n = 3200$ and those of Table 5. Of these, 16 are within ± 1.96; the two exceptions are among the deviations discussed above.

Figure 2 shows the estimates as a function of n. For the four families at $p = 5$, the standard errors remain roughly constant as n grows. One of the twelve points below $n = 4{,}000$ lies outside ± 1.96 standard errors (the scale mixture at $n = 1{,}000$, $z = -2.21$), while the neighboring points at $n = 500$ and $n = 2{,}000$ are within 0.6 standard errors. With twenty-four family checks, an isolated deviation of this size is compatible with Monte Carlo variation.

Table 5: Further verification of Theorem 2: four covariate families at $p \in \{2, 5, 8\}$, $n = 4{,}000$, $B = 20{,}000$. Targets are exact; estimates are the control-variate estimator with Monte Carlo standard errors in parentheses; the last column is the standard error of the raw estimator on the same replications.

Family	p	D_p (exact)	Estimate	z	Raw SE
Gaussian	2	-1.47	-1.31 (0.20)	$+0.82$	2.27
Gaussian	5	-2.94	-2.57 (0.42)	$+0.90$	3.22
Gaussian	8	-4.41	-3.50 (0.62)	$+1.48$	4.00
Scale mixture	2	14.81	13.33 (1.06)	-1.40	8.14
Scale mixture	5	25.56	26.51 (1.83)	$+0.52$	10.39
Scale mixture	8	36.30	35.37 (2.74)	-0.34	12.85
Rademacher	2	-9.80	-9.51 (0.29)	$+1.02$	2.97
Rademacher	5	-15.68	-16.65 (0.62)	-1.55	6.20
Rademacher	8	-21.56	-20.94 (0.94)	$+0.67$	8.13
Quadratic misspec.	2	4.81	4.75 (0.35)	-0.17	3.12
Quadratic misspec.	5	13.68	14.10 (0.96)	$+0.44$	4.99
Quadratic misspec.	8	22.56	22.19 (1.72)	-0.21	6.77

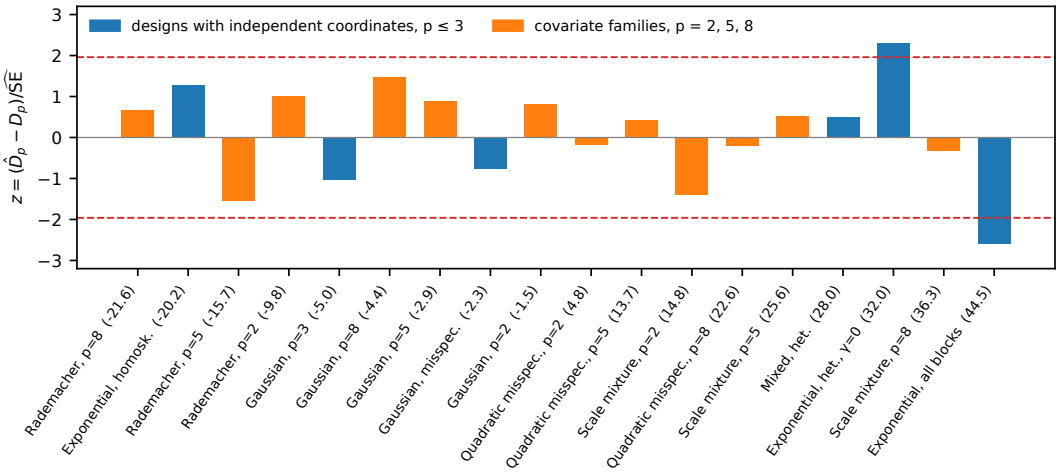


Figure 1: z-statistics $(\hat{D}_p - D_p)/\widehat{\mathrm{SE}}$ of the control-variate estimates for the designs of Table 3 ($n = 3{,}200$) and Table 5 ($n = 4{,}000$), sorted by D_p (in parentheses). Dashed lines mark ± 1.96.

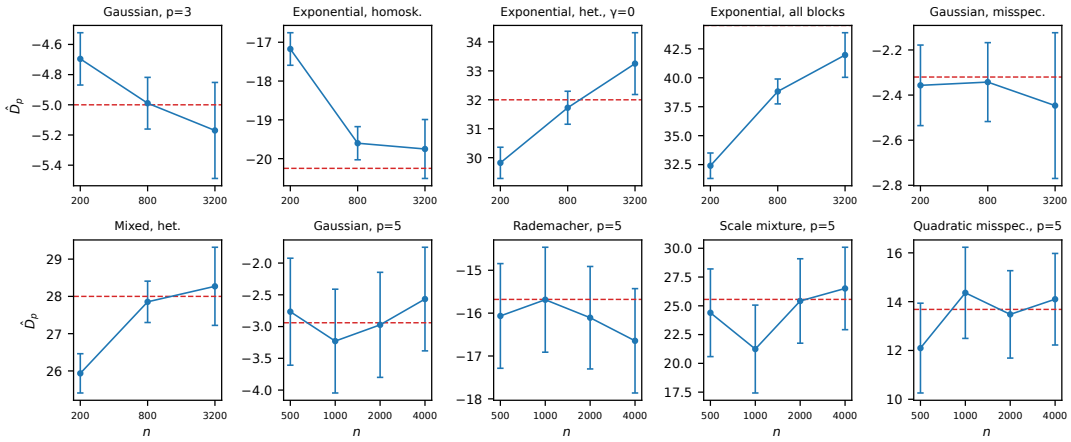


Figure 2: Control-variate estimates of D_p against n (log scale) with ± 1.96 standard-error bars; the dashed line is the exact D_p. The first six panels are the designs of Table 3 ($n = 200$, 800, 3,200); the last four are the families of Table 5 at $p = 5$ ($n = 500$ to 4,000).

The CRT correction and the bias decomposition. We check Theorem 9 separately by estimating the n^{-2} constant under both balanced CRT and Bernoulli(1/2) assignment on the same designs (`verification/crt_correction_certificate.py`). In the Gaussian design, the estimated gap between the two designs is 4.3 ± 0.6 at $n = 200$ and 6.3 ± 1.6 at $n = 800$, compared with the predicted $4G = 5$. In the exponential-covariate design, the corresponding gaps are 3.0 ± 1.1 and 8.6 ± 3.0. All four comparisons have $|z| < 2$.

This check uses the raw statistic because the control variate is derived for the two-independent-samples representation of the CRT, so its power is limited. The ablation in Table 4 provides a sharper test: there, the Bernoulli constant is rejected by as many as thirty standard errors.

We also check Remark 3 by estimating the biases of both estimators directly (`simulations/run_bias.py`). With 10^6 replications, at $n = 800$ all six estimated biases (REG and IREG in three designs) are within 0.8 Monte Carlo standard errors of their leading-order predictions. At $n = 200$, the all-blocks design differs by about 11% of the predicted REG bias ($z = 9.3$) and by 5% of the predicted IREG bias ($z = 3.5$); by $n = 800$, both differences are negligible at the reported Monte Carlo precision ($z = 0.4$ and $z = -0.5$), consistent with the $O(n^{-2})$ remainder in Lemma 13.

C Range of validity of the second-order approximation

Theorem 2 is an asymptotic result. A separate question is when the approximation

$$1 + \Delta/n$$

is accurate for the finite-sample MSE ratio. We examine this in the scalar lognormal family of the introduction, with $\sigma(X) = 1 + X$, now with a treatment-effect slope of 0.4 fixed while $X \sim \text{LogNormal}(0, s^2)$ varies. As s increases, the covariate tails become heavier and the higher moments entering Δ grow rapidly.

For $s = 0.8$ (skewness 3.7), the simulated excess MSE is 75% of the predicted excess Δ/n at $n = 500$ and 85% at $n = 2{,}000$; for $s = 0.6$, the corresponding figures are 91% and 94%. The

Table 6: Simulated MSE ratio $\widehat{\mathrm{MSE}}_{\mathrm{IREG}}/\widehat{\mathrm{MSE}}_{\mathrm{REG}}$ versus the second-order prediction $1+\Delta/n$, for $X \sim \mathrm{LogNormal}(0, s^2)$, $\sigma(X) = 1 + X$, treatment-effect slope 0.4. Monte Carlo standard errors of the simulated ratios are given in parentheses in units of the last reported digit. The introductory example is this family at $s = 1.1$ with no treatment-effect heterogeneity; the heavy-tailed example ($s = 1.8$, $\sigma(X) = 1 + 1.5X$, slope 0.3) is a separate design.

			$n = 500$		$n = 2000$		$n = 8000$	
s	skew Z	Δ	sim.	pred.	sim.	pred.	sim.	pred.
0.6	2.3	4	1.0072 (1)	1.0079	1.0019 (1)	1.0020	1.0004 (1)	1.0005
0.8	3.7	12	1.0184 (3)	1.0243	1.0052 (2)	1.0061	1.0014 (1)	1.0015
1.0	6.2	41	1.0429 (7)	1.0827	1.0137 (4)	1.0207	1.0043 (2)	1.0052
1.2	11.2	166	1.0923 (15)	1.3323	1.0347 (11)	1.0831	1.0127 (8)	1.0208
1.5	33.5	2025	1.2517 (51)	5.0503	1.1163 (44)	2.0126	1.0534 (36)	1.2531

approximation deteriorates as the tails become heavier. At $s = 1.5$, for example, the predicted ratio is four times the simulated ratio at $n = 500$. We do not interpret this as a universal threshold: the point is that a large deficiency can coexist with substantial finite-sample error in the second-order approximation.

The heavy-tailed example illustrates the same phenomenon more sharply. With $s = 1.8$, $\sigma(X) = 1 + 1.5X$, and treatment-effect slope 0.3, we have $\Delta \approx 3.9 \times 10^4$. The second-order prediction is then 78.5, 20.4, and 5.8 at $n = 500$, 2,000, and 8,000, compared with simulated MSE ratios of 1.68, 1.34, and 1.19. The sign of the comparison is correct, but $1 + \Delta/n$ is no longer a useful numerical approximation to the MSE ratio. In the same design, $R^2_\tau \approx 0.009$, and the simulated coverage of IREG intervals without the plug-in correction is 85.6% with HC1, 88.9% with HC2, and 94.0% with HC3, compared with 94.4% for REG+HC1 and 96.7% for REG+HC3; with the plug-in, IREG+HC1 covers 94.3% (Table 1).

Tail sensitivity. The largest discrepancies occur when the moments entering Δ are dominated by the far tail. Table 7 recomputes $1 + \Delta/n$ at $n = 500$ after truncating the lognormal law at several high quantiles; the truncated moments are available in closed form. Truncating at the $1 - 10^{-4}$ quantile reduces Δ by one to three orders of magnitude, showing how strongly the deficiency depends on extreme covariate values. Even truncation at the $1 - 10^{-8}$ quantile still gives predictions several times larger than the simulated ratios.

In these designs, the problem begins already at first order. For the same family without treatment-effect heterogeneity ($\gamma = 0$, the introductory design at other values of s), at $s = 0.6$ the simulated standard deviation of $\hat{\tau}_{\mathrm{REG}}$ is close to the first-order benchmark $\sqrt{V^*/n}$, 0.21 versus 0.21. At $s = 1.5$ and $s = 2$, the corresponding values are 0.72 versus 0.88 and 2.46 versus 4.90, respectively. Thus the heavy-tailed designs are outside the regime where the perturbative expansion gives an accurate quantitative approximation. Monte Carlo error is a secondary issue in the heaviest rows, where rare draws contribute substantially to the simulated MSE. The sign of the REG–IREG comparison agrees with the second-order calculation in every design.

D Designs of the inference simulations

The designs of Tables 1 and 2 all have $n = 500$, balanced complete randomization and 20,000 replications, and every interval is computed on the same simulated datasets. The scalar log-

Table 7: Second-order prediction $1 + \Delta/n$ at $n = 500$ for the full lognormal law and for its truncation at the q quantile, against the simulated MSE ratio (Monte Carlo standard error in parentheses, in units of the last digit). The last row is the family of Table 6 at $s = 2$ without treatment-effect heterogeneity ($\gamma = 0$).

Design	Simulated ratio	$1 + \Delta/n$ full law	$q = 1 - 10^{-4}$	$q = 1 - 10^{-6}$	$q = 1 - 10^{-8}$
Heavy-tailed example ($s = 1.8$)	1.68 (3)	78.5	1.90	7.94	26.6
Table 6, $s = 1.5$	1.25 (1)	5.05	1.36	2.44	3.81
$\gamma = 0$, $s = 2$	2.00 (6)	672.1	2.45	17.3	97.4

normal designs extend the family of the introduction:

$$p = 1, \qquad X_i \sim \text{LogNormal}(0, s^2), \qquad u_i \sim N(0,1),$$
$$Y_i(0) = 0.9X_i + \sigma(X_i)u_i, \qquad Y_i(1) = Y_i(0) + 1 + gX_i,$$

with $\sigma(X) = 1 + X$ except in the $s = 1.8$ design, where $\sigma(X) = 1 + 1.5X$, and

$$(s, g) \in \{(1.2, 0.3), (1.5, 0.1), (1.5, 0.5), (1.8, 0.3), (2, 0.1), (1.5, 1), (1.5, 3)\}.$$

The remaining designs have $p = 5$, potential outcomes that share their noise, and

$$Y_i(0) = m(X_i) + \sigma(X_i)u_i, \qquad Y_i(1) = Y_i(0) + 1 + \gamma^* X_{i1}, \qquad u_i \sim N(0,1),$$

with $m(x) = 0.9 \sum_j x_j$ unless stated. They differ in the covariate law, the noise scale and the slope γ^*:

- *Gaussian*: $X_i \sim N(0, I_5)$, $\sigma \equiv 1$ and $\gamma^* = 0.3$; the large-heterogeneity Gaussian row instead has $X_i \sim N(0, \Sigma)$ with $\Sigma_{jk} = 0.5^{|j-k|}$ and $\gamma^* = 3$.
- *Misspecified, $G = 0$*: $X_i \sim N(0, \Sigma)$ with $\Sigma_{jk} = 0.5^{|j-k|}$, $\sigma \equiv 1$, $\gamma^* = 0$, and $m(x) = 0.9 \sum_j x_j + 0.20(x_1^2 - 1) + 0.15 x_1 x_2$, so that the working model is misspecified in both arms.
- *Skewed, het., $G = 0$*: independent coordinates, the first a centered LogNormal$(0, 1.1^2)$ and the others standard normal, mixed by the Cholesky factor of the matrix $0.35^{|j-k|}$ and then standardized to unit variances; $\sigma(X) = 1 + 0.5|X_{i1}| + 0.25\|X_i\|$ and $\gamma^* = 0$.
- *Heavy-tailed, het.*: the same construction with LogNormal$(0, 1.5^2)$ and $\sigma(X) = 1 + 0.75|X_{i1}| + 0.25\|X_i\|$; $\gamma^* = 0.5$ in the small-heterogeneity row and $\gamma^* = 3$ in the large-heterogeneity row.
- *Rare extreme values*: $X_{i1} = \sqrt{(1-\rho)/\rho}$ with probability $\rho = 0.02$ and $-\sqrt{\rho/(1-\rho)}$ otherwise, $X_{i2}, \ldots, X_{i5} \sim N(0,1)$, $\sigma \equiv 1$ and $\gamma^* = 1$. With 250 units per arm, an arm contains no rare unit with probability about 0.6%, in which case the interacted design is rank deficient and IREG gives no interval (1.3% of replications), and exactly one rare unit, which then has leverage one in its arm's fit, with probability about 3%.